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**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

Seventh Semester B.Tech Degree Examination December 2022 (2019 scheme)

**Course Code: MCN401**

**Course Name: INDUSTRIAL SAFETY ENGINEERING**

**Max. Marks: 100**

**Duration: 3 Hours**

**PART A**

*Answer all questions, each carries 3 marks.*

Marks

- |    |                                                                                                           |     |
|----|-----------------------------------------------------------------------------------------------------------|-----|
| 1  | Explain any three unsafe acts which are responsible for accidents in industries.                          | (3) |
| 2  | List any important six responsibilities of worker/workmen towards the safety measures in an organization. | (3) |
| 3  | What is respiratory protective equipment (RPE)? Explain the features of any one type of RPE.              | (3) |
| 4  | How does frequency rate and incidence rate support safety analysis?                                       | (3) |
| 5  | List any important six safety practices used with ladders in construction sites.                          | (3) |
| 6  | Explain any important four safety practices in excavation works.                                          | (3) |
| 7  | Mention any four potential hazards associated with wire rope used for material handling.                  | (3) |
| 8  | Discuss the key elements of a hearing conservation program.                                               | (3) |
| 9  | Explain the hierarchy of control of chemical hazards.                                                     | (3) |
| 10 | Why material safety data sheet is mandatory for chemical products?                                        | (3) |

**PART B**

*Answer any one full question from each module, each carries 14 marks.*

**Module I**

- |    |                                                                                                  |     |
|----|--------------------------------------------------------------------------------------------------|-----|
| 11 | a) With suitable schematics, describe the different types of safety organization.                | (6) |
|    | b) Explain the responsibilities of safety officer in the implementation of safety in industries. | (8) |

**OR**

- |    |                                                                                                      |      |
|----|------------------------------------------------------------------------------------------------------|------|
| 12 | List the various accident causation theories and explain any two in detail with relevant schematics. | (14) |
|----|------------------------------------------------------------------------------------------------------|------|

**Module II**

- |    |                                                                                                         |     |
|----|---------------------------------------------------------------------------------------------------------|-----|
| 13 | a) Explain the benefits of good housekeeping? Also, explain the five 'S' (5 S) concept in housekeeping. | (9) |
|    | b) With suitable sketches explain the important functions of any two PPE used for eye protection.       | (5) |

**OR**

- 14 a) What are the major objectives of a work permit system in a hazardous work site? (5)
- b) Discuss any five potential hazards associated with hot works. Hence, mention any eight safety measures referred to hot work permits. (9)

**Module III**

- 15 a) You are appointed as a safety manager for a demolition work of a 10-storey building in a congested residential area. What are the safety measures that you need to ensure for the completion of the work? (8)
- b) Discuss the major ergonomic hazards associated with construction industries. (6)

**OR**

- 16 a) What are the safety precautions to be taken before entering a confined space? (9)
- b) Discuss some important aspects of construction safety provisions in National Building Code. (5)

**Module IV**

- 17 a) With suitable sketches explain the operation of any two types of safety guards suitable for industrial applications. (8)
- b) Explain the potential hazards associated with grinding operations. (6)

**OR**

- 18 a) Discuss the safety issues associated with gas welding operations. (8)
- b) Discuss about the handling capacity assessment for lifting process associated with manual material handling. (6)

**Module V**

- 19 a) Explain the need for a Preliminary Hazard Analysis in a hazardous industry. (6)
- b) Explain the important features and functions of any four different types of fire extinguishers. (8)

**OR**

- 20 a) What is the significance of Hazard and Operability Analysis? How do you conduct a HAZOP analysis? (8)
- b) Discuss about different types of chemical hazards with suitable examples. (6)

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